

Multiple Tree Model Integration for Transportation Mode Recognition

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ABSTRACT

The team RY presents a solution for Sussex-Huawei Locomotion-Transportation (SHL) recognition challenge, which aims at differentiating eight transportation modes with mobile phone signal sensor data in this paper. This study first extracted a set of reasonable and discriminative features after data-preprocessing. Then, decision tree bagging, random forest, lightGBM are trained separately as basic models, whose predictions are integrated and afterward smoothed. The method gets 0.65 accuracy score on validation dataset.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; *Simulation environments*; • **General and reference** → Cross-computing tools and techniques.

KEYWORDS

Transportation Mode Recognition; LightGBM; Random Forest; Bagging; Ensemble

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1 INTRODUCTION

Transportation Mode Recognition (TMR) based on mobile devices is a popular subject these years. As a branch of Motion Mode Recognition (MMR), TMR could be used in various fields such as transportation planning, physical activity monitoring etc. Meanwhile, with the improvement of communication technology, much data can be easily collected by sensors and signal receivers built in mobile phones. Based on Sussex-Huawei Locomotion Dataset [6, 13], the challenge this year is to recognize 8 transportation modes (Still, Walking, Run, Bike, Car, Bus, Train, Subway) using signal data collected by mobile phones, including GPS, WiFi and Cells data.

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Different from previous challenges [11, 12, 14], this time, signal data is focused instead of motion data collected directly by mobile devices tied to participants. Main difficulties needed to be conquered are listed below: 1) There is probably no signal or multiple entries within a timestamp. Time gaps among different signal receivers are also needed to be balanced; 2) Fewer semantic features can be attracted from signal data to classify different transportation modes compared with motion data. However, there are also advantages of signal data. For example, there are fewer individual differences in each transportation modes because signals collected by mobile phones are more based on position, and those features are not user-dependent. Furthermore, if it is possible to classify motion states accurately using signal data only, it can be a rather power-efficient method which can be used online.

In short, in SHL recognition challenge this year, we aim to settle transportation mode recognition task using signal data received by mobile phones. Inspired by previous work illustrated in article [1, 5, 9], the method of this study can be divided as 5 main steps: data pre-processing, feature extraction, modeling, model integration and data post-processing.

2 DATASET DESCRIPTION

Sussex-Huawei Locomotion Dataset is used to accomplish the challenge. 59 days are recorded by User1 in training data. Validation data is a mixture of data collected by User2 and User3 in 6 days. There are 4 sources of information provided (note that timestamps are available for each source):

- **Location:** Latitude, Longitude, Altitude and Accuracy of this location. Based on this file, user locations in UTM coordinate are available. Then speed, acceleration are calculated accordingly.
- **GPS:** Number of GPS entries. For each entry, we have satellite ID, SNR (Signal-to-Noise Ratio), Azimuth, Elevation. Among them, SNR is the indicator of GPS signal strength and accuracy, and therefore informative.
- **WiFi:** Number of WiFi entries. For each entry, BSSID, SSID, RSSI, Frequency in MHz, Capabilities are recorded. Among them, RSSI indicates signal strength, while Frequency means 2.4 GHz channel or 5 GHz channel.
- **Cells:** Number of Cells signal entries. For each entry, informative features include signal level, signal strength in dBm and whether this entry is registered.

Apart from the location file, the data is not neatly structured. In other words, the data lengths of different timestamps are not aligned due to different signal entry numbers. Aggregation Operations should be taken accordingly.

3 DATA PROCESSING & FEATURE EXTRACTION

3.1 Timestamp & Missing Value

Although the sampling rate is roughly 1 Hz, timestamps of different sensors do not match with each other totally. The unit of time recorded is actually ms, and it is rather common that not all sensors have data collected in a second. Consequently, many NaN values wait to be handled. A 0-1 feature is added for each source to discriminate between no signal received (take WiFi data as an example, it means that the number of WiFi entries equals to 0) and the sensor is not in use at that specific timestamp (the number of WiFi is NaN, which is generated by merging dataframes). Then, number 0 is used to replace all other NaNs. In this way, when the models are all tree-based, there is no need to worry about confusing original 0 value and 0 replacing NaN.

3.2 Feature Extraction

Features extracted from the Sussex-Huawei Locomotion Dataset could be classified into point features and window features. Both of them are in statistical domain. Table 1 lists all hidden features taken into account before further feature selection.

Table 1: Overall Feature Extraction Method

Source File	Method	Basic Feature
Location	original value, window std	speed, acceleration
GPS	aggregation: min, max, mode, mean, std	SNR
WiFi	aggregation: min, max, mode, mean, std	RSSI, frequency
Cells	aggregation: min, max, mode, mean, std	isRegistered, dBm, asuLevel, level

With the help of UTM transformation illustrated in reference articles [3, 4], the speed and acceleration of each timestamp can be calculated. Notice that the timestamps are not totally continuous, so to ensure the accuracy of derived features, Δt -based features like speed and acceleration are derived only for continuous timestamps. These features of other timestamps are replaced by 0. Moreover, since the variance of speed changes greatly among different transportation modes, corresponding window features should be also derived. After roughly testing window sizes from 30s to 300s with the step length of 30s and considering the interpretability, 120s window size is finally taken.

For GPS, WiFi and Cells data, aggregation method listed in Table 1 are used. In this way, data line with different entry numbers are unified. Minimum, maximum value and standard variance indicate difference among entries, while mode and mean reflect the average level of each feature.

Notice that not all methods are applied to every basic feature. Only reasonable aggregated features are adopted. For instance, the mean of isRegistered in Cells file can be understood as register ratio, but the mode of it does not make sense. The same overall semantic feature selection logic applies to other variables as well.

3.3 Feature Selection

If all reasonable methods mentioned above are used, the model will suffer from variable redundancy. Therefore, a series of descriptive statistics are necessary to screen out features that do not apparently differentiate among different transportation modes. In this step, possible data transformations are also explored. As a result, a set of discriminative features are ready for training. Figure 1 shows speed differences among states. A log transformation is applied here to handle right-skew problem.

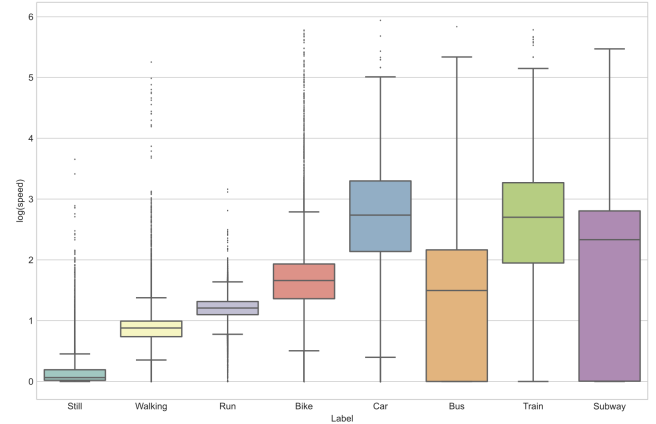


Figure 1: Log(speed) Difference Among States

From descriptive diagrams like Figure 1, it is referable that speed feature is potential for discriminating among still, walking, run and bike. But the speed feature will confuse car with train, bus with subway. In the same way, advantages and disadvantages exist for other features.

So far, 32 semantic reasonable and discriminative features are selected and properly transformed to train the models as multi-class classifier. These features are listed in Table 2.

4 PIPELINE

Many multi-class classifiers are available for this task. Machine learning methods are extensively used by previous researchers on TMR task. Single machine learning model includes SVM [10], decision tree [15], regression method. Ensemble learning model includes stacking, bagging and boosting. There are also many researches about deep learning method [2, 7, 16]. Typically, deep neural networks perform better than traditional models in accuracy after careful tuning. However, the time cost to train and tune the neural network outweighs the accuracy gain.

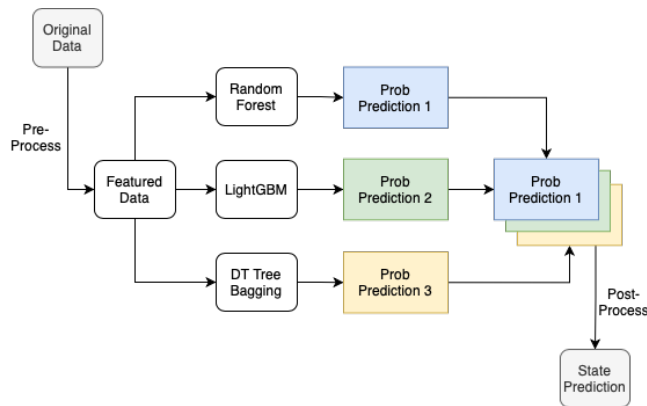
In this study, three traditional but time-efficient tree-based models (decision tree bagging, random forest, lightGBM) are used and properly combined. Tree-based models are chosen to handle confusing 0 value of data illustrated in Section 3.1. Model integration strategy is stated in Section 4.2.

4.1 Basic Models

The basic models applied in this study are all ensemble-based. They themselves already consist sets of trees with different constructing and combination strategy.

Table 2: 32 Selected Features

File	Original Features	Derived Number	Method
Location	accuracy	1	original value
	speed	4	log transformation, window std, log of window maximum, std of window maximum
	acceleration	2	original value, window std
GPS	number	1	entry number
	snr	4	mean, min, max, std
WiFi	number	1	entry number
	rss	5	mode, mean, min, max, std
	frequency	1	mean
Cells	number	1	entry number
	isRegistered	1	mean
	asuLevel	4	mean, min, max, std
	dBm	4	mean, min, max, std
	level	3	mean, min, max

**Figure 2: Process flow chart**

4.1.1 Decision Tree Bagging. Bagging is one of the simplest ensemble-based algorithm to enhance prediction accuracy of single models. DT bagging model constructs many decision trees with different sets of randomly sampled data.

4.1.2 Random Forest. Random forest is actually the enhanced version of DT bagging model with the introduction of randomness in selecting variables. It reduces correlation among sub-trees.

4.1.3 LightGBM. LightGBM is the faster version of XGBoost, which is one of the popular gradient boosting models. Instead of setting goals of all trees to be the same, gradient boosting methods construct the next sub-tree to learn residuals left by the last sub-tree. LightGBM shares the same principle idea, but faster than others.

4.2 Ensemble Method and Smoothing

For each basic model, a probability prediction can be calculated. First, an element-wise addition is applied to combine 3 probability predictions into one as shown in Figure 2. Normalizing by row is not necessary here, because the sum of each row does not affect where the largest value appears. Second, the position of the largest value is located. Then a raw prediction label series is gotten.

However, it does not make sense to have a prediction series as $\{...Run, Run, Run, Subway, Run, Run, ...\}$. In other words, point predictions may not be accurate, but once smoothed with predictions of surrounding points, a higher accuracy may be achieved. In actual operation, for each timestamp, original point predictions are replaced by mean values within the window. 120s is again chosen as the window size by experience.

Finally, further rules are in place to ensure that predictions are smooth. The predicted label whose duration is less than 5s is replaced by the previous stable prediction. The threshold is set low to prevent collapsing errors.

5 RESULTS

5.1 Experimental Setup

The Experiments were taken locally on a 2 GHz Inter Core i5 CPU, with 16 GB memory. In training and tuning process, original training dataset is randomly split by 8:2 to be real training data and temporal test data. In real training dataset, there are 784,421 lines with 32 features. Table 3 shows the time consumed, multi-class accuracy of temporal test data, multi-class accuracy of validation data of each model.

Generally, validation dataset is used to tune models and test dataset is used to get scores. However, since data types of Sussex-Huawei Locomotion Dataset (training, validation and test) are specified and there are no true labels in test dataset, the temporal test data is virtually used as conventional validation data, and validation dataset is used as conventional test data. Notice the names to avoid confusion about the results.

Table 3: Overall Feature Extraction Method

Model	Time (s)	Acc (Test)	Acc (Val)
DT Bagging	27.87	0.69	0.64
Random Forest	32.41	0.77	0.60
LightGBM	31.88	0.94	0.59

It can be observed that there is no significant difference of training time. On the test data split out from original training dataset, LightGBM outperforms others and DT Bagging reaches the lowest accuracy. However, the rank reverses on validation data. Recall that validation dataset is a mixture of User2 and User3 while training dataset (from which the temporal test data split) is only composed of data from User1. The accuracy gap between datasets can not be blamed as overfitting because the temporal test data is currently not used in training as well. It could be inferred that there are still individual differences. Although signal data is less user-dependent,

other variables like speed and acceleration are still included in features shown in Table 2. Model integration is tried to mitigate the difference.

5.2 Integrated Results

In this part, all original training dataset is used to train the model. The following results are all about validation dataset. Apply integration strategies illustrated in Section 4.2, the confusion matrix is shown in Figure 3, and statistical scores are listed in Table 4.

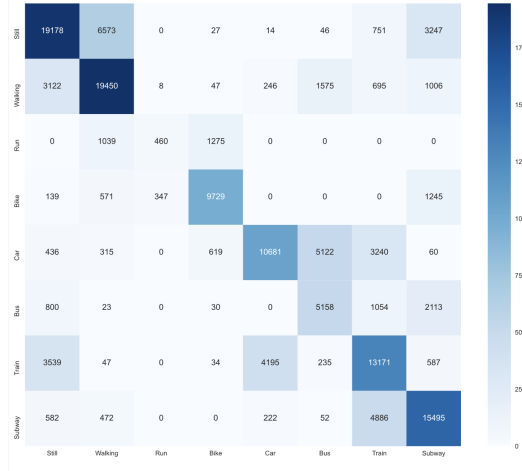


Figure 3: Confusion Matrix on Validation Dataset

Table 4: Prediction Score Detail

	Precision	Recall	F1-Score	Support
Still	0.68	0.64	0.66	29836
Walking	0.68	0.75	0.72	26149
Run	0.61	0.15	0.25	2774
Bike	0.82	0.81	0.82	12031
Car	0.72	0.55	0.62	20473
Bus	0.43	0.55	0.48	9178
Train	0.57	0.61	0.59	21808
Subway	0.65	0.71	0.68	21709
accuracy			0.65	143958
macro avg	0.65	0.60	0.60	143958
weighted avg	0.66	0.65	0.65	143958

The model performs best in identifying *Bike* state with f1-score 0.82, worst in *Run* state with f1-score 0.25. Statistical scores of predictions before smoothing (illustrated in Section 4.2) and after that are almost the same. But this smoothing step really makes sense when plotting prediction series chronologically as shown in Figure 4 and Figure 5

Blue lines indicate predicted state series and black lines show the ground truth. It can be easily observed that after taking the mean of predicted probability matrix in a window and do rule-based smoothing, the prediction state series is stabler.

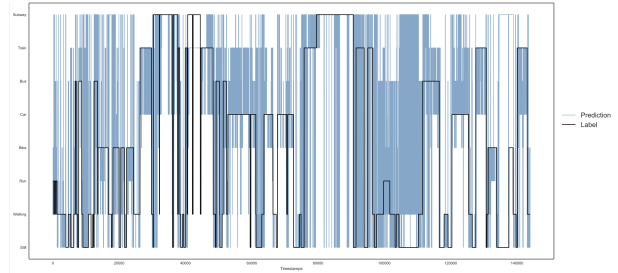


Figure 4: Prediction Series Before Smoothing

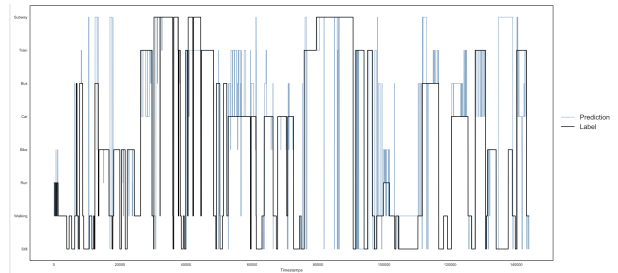


Figure 5: Prediction Series After Smoothing

6 DISCUSSION

This study proposes an ensemble method of tree-based models to accomplish SHL recognition challenge. Reasonable and discriminative features are extracted before training. After prediction, post-processing is applied to ensure the local consistency of prediction series, which enhances the accuracy as well. However, there are still some drawbacks in this study to be further explored.

- More semantic features should be derived with specialized knowledge in signal field. Currently, only conventional statistical methods are applied.
- Similar data contamination exists between training data and temporal test data defined in Subsection 5.1. Because these two datasets are split randomly from original training dataset, it is often the case that data of time t belongs to training data, whereas data of time $t + 1$ belongs to temporal test data. Features of adjacent timestamps are similar, which may attribute to high scores of temporal test data shown in Table 3. An alternative splitting method has already tried, which means splitting data directly into two continuous parts. However, only two transportation modes are included in temporal test data, so test scores are not reliable in this way.
- All three basic models are tree-based. Although their structures are not totally the same and tree-based models generally perform well in classifying tasks, the inherent consistency of their predictions, which can be wrong, is still a problem. For example, three models are likely to agree on predicting a series of points as *Bike* states when the true labels are all *Run* states. Models of other structures are needed to be explored to truly enhance ensemble effect.

- A more intelligent post-processing method is needed. A hard threshold is set with the strategy illustrated in Subsection 4.2, which is apparently not flexible enough for different transportation modes.
- The model can not handle individual differences well. If data of more users can be get as training dataset, there is no problem about the model. But if data of only one user constitutes the training data like Sussex-Huawei Locomotion Dataset, more tricks should be used. One possible way is to tag individuals using another feature. However, it only works for small individual number, which is not the case in real application. In this study, original training dataset and validation dataset are combined to train the final model used for predicting test data (no label provided). It is not a typical use of datasets, but it can relieve individual difference impact considering test data is a mixture of User2 and User3 as validation dataset (user constitution of datasets are explained in Section 2).

7 CONCLUSION

In short, this method is time efficient and reasonable though with disadvantages stated in Section 6. Three tree-based models (decision tree bagging, random forest, lightGBM) are properly integrated as the final model in this study. The method gets 0.65 accuracy score on validation dataset, which can set a baseline for the problem. The recognition result for the testing dataset will be presented in the summary paper of the challenge [8].

REFERENCES

- [1] Huthaifa I. Ashqar, Mohammed H. Almannaa, Mohammed Elhenawy, Hesham A. Rakha, and Leanna House. 2019. Smartphone Transportation Mode Recognition Using a Hierarchical Machine Learning Classifier and Pooled Features From Time and Frequency Domains. *IEEE Transactions on Intelligent Transportation Systems* 20, 1 (2019), 244–252. <https://doi.org/10.1109/TITS.2018.2817658>
- [2] Francesco Delli Priscoli, Alessandro Giuseppi, and Federico Lisi. 2020. Automatic Transportation Mode Recognition on Smartphone Data Based on Deep Neural Networks. *Sensors* 20, 24 (2020). <https://doi.org/10.3390/s20247228>
- [3] Mostafa Elhoushi, Jacques Georgy, Michael Korenberg, and Aboelmagd Noureldin. 2014. Broad motion mode recognition for portable navigation. In *Proceedings of the 27th International Technical Meeting*, Vol. 2. 27th International Technical Meeting of the Satellite Division of the Institute of Navigation, ION GNSS 2014, New York, NY, USA.
- [4] Mostafa Elhoushi, Jacques Georgy, Michael Korenberg, and Aboelmagd Noureldin. 2014. Robust motion mode recognition for portable navigation independent on device usage. In *Record - IEEE PLANS, Position Location and Navigation Symposium*. Proceedings of IEEE/ION PLANS 2014, Monterey, CA. <https://doi.org/10.1109/PLANS.2014.6851370>
- [5] Tao Feng and Harry J.P. Timmermans. 2013. Transportation mode recognition using GPS and accelerometer data. *Transportation Research Part C: Emerging Technologies* 37 (2013), 118–130. <https://doi.org/10.1016/j.trc.2013.09.014>
- [6] H. Gjoreski, M. Ciliberto, L. Wang, F.J.O. Morales, S. Valentin S. Mekki, and D. Roggen. 2018. The University of Sussex-Huawei locomotion and transportation dataset for multimodal analytics with mobile devices. *IEEE Access* 6 (2018), 42592–42604.
- [7] Jeya Vikranth Jeyakumar, Eun Sun Lee, Zhengxu Xia, Sandeep Singh Sandha, Nathan Tausik, and Mani Srivastava. 2018. Deep Convolutional Bidirectional LSTM Based Transportation Mode Recognition. In *Pervasive and Ubiquitous Computing and Wearable Computers* (Singapore, Singapore) (UbiComp '18). Association for Computing Machinery, New York, NY, USA, 1606–1615. <https://doi.org/10.1145/3267305.3267529>
- [8] L. Wang and M. Ciliberto and H. Gjoreski and P. Lago and K. Murao and T. Okita, and D. Roggen. 2021. Locomotion and transportation Mode Recognition from GPS and radio signals: Summary of SHL Challenge 2021. In *Proceedings of the 2021 ACM International Joint Conference and 2021 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers*. Association for Computing Machinery, New York, NY, USA.
- [9] Sasank Reddy, Min Mun, Jeff Burke, Deborah Estrin, Mark Hansen, and Mani Srivastava. 2010. Using Mobile Phones to Determine Transportation Modes. *ACM Trans. Sen. Netw.* 6, 2, Article 13 (March 2010), 27 pages. <https://doi.org/10.1145/1689239.1689243>
- [10] Jian Shen, Haihang Jiang, Fei Yang, and Zhenxing Yao. 2019. Trip mode recognition using smartphone sensor data under different sampling frequencies. *Web Intelligence* 17, 2 (2019), 151–160. <https://doi.org/10.3233/WEB-190409>
- [11] L. Wang, H. Gjoreski, M. Ciliberto, P. Lago, K. Murao, T. Okita, and D. Roggen. 2019. Summary of the Sussex-Huawei Locomotion-Transportation Recognition Challenge 2019. In *Proceedings of the 2019 ACM International Joint Conference and 2019 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers* (London, United Kingdom) (UbiComp-ISWC '19 Adjunct). Association for Computing Machinery, New York, NY, USA, 849–856. <https://doi.org/10.1145/3341162.3344872>
- [12] L. Wang, H. Gjoreski, M. Ciliberto, P. Lago, K. Murao, T. Okita, and D. Roggen. 2020. Summary of the Sussex-Huawei Locomotion-Transportation Recognition Challenge 2020. In *Proceedings of the 2020 ACM International Joint Conference and 2020 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers* (Virtual Event, Mexico) (UbiComp-ISWC '20). Association for Computing Machinery, New York, NY, USA, 351–358. <https://doi.org/10.1145/3410530.3414341>
- [13] L. Wang, H. Gjoreski, M. Ciliberto, S. Mekki, S. Valentin, and D. Roggen. 2019. Enabling reproducible research in sensor-based transportation mode recognition with the Sussex-Huawei dataset. *IEEE Access* 7 (2019), 10870–10891.
- [14] L. Wang, H. Gjoreskia, K. Murao, T. Okita, and D. Roggen. 2018. Summary of the Sussex-Huawei Locomotion-Transportation Recognition Challenge. In *Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers* (Singapore, Singapore) (UbiComp '18). Association for Computing Machinery, New York, NY, USA, 1521–1530. <https://doi.org/10.1145/3267305.3267519>
- [15] Shuangquan Wang, Canfeng Chen, and Jian Ma. 2010. Accelerometer Based Transportation Mode Recognition on Mobile Phones. In *2010 Asia-Pacific Conference on Wearable Computing Systems*. IEEE, Shenzhen, China, 44–46. <https://doi.org/10.1109/APWCS.2010.18>
- [16] Hong Zhao, Chunming Hou, Hala Alrobassy, Xiangyan Zeng, and Giovanni Pau. 2019. Recognition of Transportation State by Smartphone Sensors Using Deep Bi-LSTM Neural Network. *J. Comput. Netw. Commun.* 2019 (Jan. 2019), 11 pages. <https://doi.org/10.1155/2019/4967261>